AWANTIKA SRIVASTAVA

AI / Machine Learning Engineer | Data Science | LLM & Computer Vision

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PROFILE SUMMARY

Machine Learning Engineer and Data Scientist with **2+ years of experience in Computer Vision, Deep Learning, and AI system development**. Experienced in designing and deploying **computer vision models for object detection, image classification, and real-time video analytics**. Skilled in **Python, PyTorch, TensorFlow, and OpenCV** for building scalable AI pipelines. Strong background in **model evaluation, image preprocessing, and large-scale dataset preparation**. Hands-on experience with **MLOps tools, cloud deployment, and production ML pipelines**. Passionate about building **AI solutions for real-world applications including healthcare and intelligent systems**.

CORE TECHNICAL SKILLS

- **Programming Languages & Tools:** Python, Numpy, Pandas, Matplotlib, Scikitlearn, C++, SQL
- **Statistics & Mathematics:** EDA, Statistical Modeling, Hypothesis Testing, Probability, Linear Optimization, Trend Analysis, Experimental Design.
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, Hyperparameter Tuning, Feature Engineering.
- **Deep Learning:** CNN, RNN, LSTM, Transformers (BERT), Transfer Learning, Model Fine-tuning, RAG, LLMs, PyTorch, TensorFlow, Keras
- **Computer Vision**: Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Image & Video Processing Pipeline, OpenCV, Bounding Boxes, Image Augmentation, Paddle OCR
- **Model Evaluation:** Precision, Recall, F1 Score, IoU, mAP, Cross Validation, A/B Testing
- **NLP & LLM:** RAG Systems, Prompt Engineering, Semantic Search, Text Processing, Transformer-based NLP, LLM Integration
- **MLOps & Deployment:** Docker, Git, Flask APIs, CI/CD, Model Monitoring, ML Pipelines
- **Cloud & Platforms:** AWS EC2, Linux, Jupyter Notebook

EXPERIENCE

Machine Learning Engineer | PPS International Pvt. Ltd. **January 2024-Present**

- Designed and deployed **computer vision and machine learning models** for real-time inference and production applications.
- Developed **deep learning pipelines using PyTorch and TensorFlow** for image classification, object detection, and anomaly detection tasks.
- Built **data preprocessing and feature engineering workflows** to process large structured and unstructured datasets.
- Implemented **OCR-based document processing systems** to extract structured information from scanned images and PDFs.
- Developed **LLM-powered applications and RAG pipelines** for document search and intelligent information retrieval.
- Designed **end-to-end ML pipelines** including data ingestion, model training, evaluation, and deployment.
- Conducted **model evaluation using metrics such as precision, recall, IoU, and mAP** to measure computer vision model performance.
- Collaborated with cross-functional teams including **data engineers and software engineers** to integrate AI systems into production environment

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- **Designed** and deployed a real-time computer vision–based ML system to **detect** unsafe driver behaviors from **continuous video streams**.
- Trained and optimized **CNN-based object detection** model (SSD **MobileNet** architecture) to perform **real-time inference** on video data.
- Implemented **end-to-end ML pipelines** covering data ingestion, **preprocessing**, **model training**, **evaluation**, and **production** inference.
- Achieved **20–25 FPS** real-time processing with **<150** ms inference latency by optimizing modelsfor deployment.
- Built and integrated a **Flask-based web interface** to **visualize** detections and automatically **recorded 30-second** event clips, **reducing** manual review effort.

Real-Time Surveillance Object Detection System | Applied ML Project

- Designed real-time surveillance system to detect and track unauthorized objects/persons in restricted zones.
- Implemented **YOLO-based object detection pipeline using OpenCV** for continuous video stream analysis.
- Applied **bounding box filtering and confidence threshold tuning** to reduce false positives.
- Implemented geometric zone mapping and spatial reasoning to detect boundary violations.
- **Optimized** detection workflow using **Non-Max Suppression (NMS)** and **IoU-based** filtering.
- **Evaluated** model **performance** using **precision**, **recall**, and **mAP** metrics.

YouTube Comments Sentiment Analyzer | link-https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Deployed transformer-based **NLP models** as **production-ready** services with **REST APIs**.
- **Fine-tuned** and served a **DistilBERT**-based **sentiment** classification model for large-scale text inference.
- **Built** and deployed an interactive **streamlit web application** to perform real-time **sentiment analysis** on YouTube comments.
- **Processed** high-volume text **data** with sub-second inference **latency** for real-time sentiment analysis.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**
Bachelor of Technology